

solution.

EXAMPLE 2.9.1

A firm manufactures two products A & B on which the profits earned per unit are Rs. 3 and Rs. 4 respectively. Each product is processed on two machines M_1 and M_2 . Product A requires one minute of processing time on M_1 and two minutes on M_2 , while B requires one minute on M_1 and one minute on M_2 . Machine M_1 is available for not more than 7 hrs. 30 mins. while machine M_2 is available for 10 hrs. during any working day. Find the number of units of products A and B to be manufactured to get maximum profit.

[P.U.B.E. (Mech.) Nov., 1996]

Formulation of Linear Programming Model

Let x_1 and x_2 denote the number of units of products A and B to be produced per day.

Objective is to maximize the profit.

i.e., maximize $Z = 3x_1 + 4x_2$.

... (2.1)

Constraints are on the time available for machines M_1 and M_2 .

i.e., for machine M_1 , $1.x_1 + 1.x_2 \leq 450$, and

for machine M_2 , $2x_1 + 1.x_2 \leq 600$,

... (2.2)

where $x_1, x_2 \geq 0$.

... (2.3)

Thus the problem is to maximize equation (2.1) subject to relations (2.2) and (2.3). This will be done graphically.

Solution of L.P. Model

The non-negativity restrictions $x_1 \geq 0$ and $x_2 \geq 0$ imply that values of the variables x_1 and x_2 can lie only in the first quadrant (x_1, x_2 plane). This is shown by shaded area of figure 2.2. Other quadrants do not satisfy the non-negativity restrictions and hence the pt. (x_1, x_2) cannot lie in them. Therefore, a number of alternatives are eliminated.

The effect of the remaining constraints can now be added to figure 2.2. This is done by plotting all the constraints with their inequality sign changed into equality sign. The direction in which each constraint holds good is then determined from the direction of the inequality and is indicated by an arrow on its associated straight line. The constraint conditions define the boundary of the region containing feasible solution.

For example, the next constraints are $x_1 + x_2 \leq 450$ and $2x_1 + x_2 \leq 600$. We plot line $x_1 + x_2 = 450$ by joining two convenient points, say (0,450) and (450,0) and line $2x_1 + x_2 = 600$ by joining points, say (0,600) and (300,0). Then any point lying on or below the line $x_1 + x_2 = 450$ satisfies the constraint $x_1 + x_2 \leq 450$. Similarly any point lying on or below the line $2x_1 + x_2 = 600$ satisfies the constraint $2x_1 + x_2 \leq 600$. This is clearly indicated by the direction of arrowheads. The shaded area in figure 2.3 satisfies both the constraints $x_1 + x_2 \leq 450$ and $2x_1 + x_2 \leq 600$ and also the non-negativity restrictions $x_1 \geq 0, x_2 \geq 0$. This area is called the *solution space* or the *region of feasible solutions*. Any point in this shaded region is a *feasible solution* to the given problem.

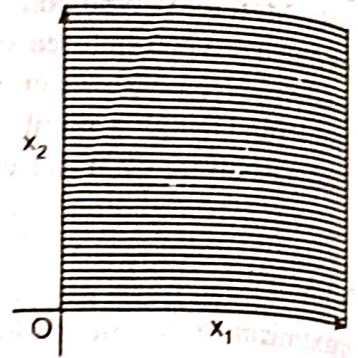


Fig. 2.2

To Find Optimal Solution

Method 1: Our problem is to find the point (or points) in the feasible region, which maximizes (or maximize) the objective function (*i.e.*, profit). To do this, we notice that when Z is made zero, equation (2.1) becomes $3x_1 + 4x_2 = 0$ and this is represented by the dotted line AB passing through origin O . Its slope *i.e.*, tangent of the angle with X -axis is $(-3/4) = -0.75^*$. As

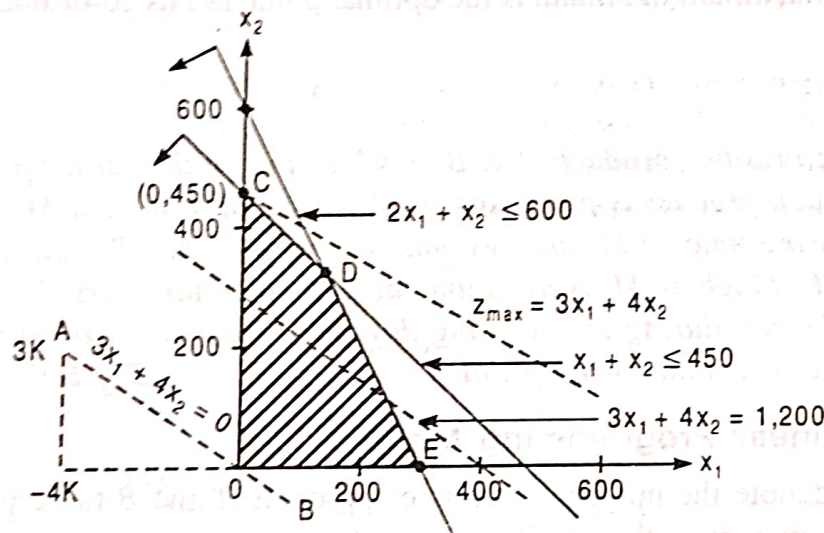


Fig. 2.3

the value of Z is increased from zero, the dotted line AB starts moving to the right, parallel to itself. Greater the value that Z can assume, more will be the company's profit. We go on drawing lines parallel to this line till the line is farthest away from the origin and passes through only one point of the feasible region. This is the point where maxima is attained. It is possible that such a line may be one of the edges of the feasible region. In that case every point on that edge gives the same maximum value of the objective function.

Another method to plot the dotted line for $Z = 3x_1 + 4x_2$ is to put $3x_1 + 4x_2 = 0$, giving $\frac{x_1}{x_2} = \frac{-4}{3}$. We plot 4 units of length on the negative side of the x_1 -axis and then 3 units of length along the x_2 -axis. The point so obtained when joined with the origin, represents the line $3x_1 + 4x_2 = 0$.

In the present example, maximum is obtained at the corner point $C(0,450)$, which means that only product B should be manufactured and 450 units of this product B should be produced.

*An easy method to plot the dotted line for the objective function $Z = 3x_1 + 4x_2$ is to assign any value to Z , say $30 \times 40 = 1,200$ and to plot the line $3x_1 + 4x_2 = 1,200$. A dotted line is then drawn through the origin parallel to this line.

The daily profit will be $Z = \text{Rs. } (0 + 4 \times 450) = \text{Rs. } 1,800$. This solution corresponding to point C (0, 450), which maximizes the objective function, is called *optimal solution*.

Method 2: The four vertices of the convex region OCDE are O (0, 0), C (0, 450), D (150, 300) and E (300, 0). Values of the objective function $Z = 3x_1 + 4x_2$ at these vertices are $Z(O) = 0$, $Z(C) = 1,800$, $Z(D) = 450 + 1,200 = 1,650$, $Z(E) = 900$.

Thus the maximum value of Z is Rs. 1,800 and it occurs at the vertex C (0, 450).

Hence the solution to the problem is

$$x_1 = 0, x_2 = 450 \text{ and } Z_{\max} = \text{Rs. } 1,800.$$

EXAMPLE 2.9.2

Find the maximum value of

$$\begin{aligned} Z &= 2x_1 + 3x_2, \\ \text{subject to } x_1 + x_2 &\leq 30, \\ x_2 &\geq 3, \\ x_2 &\leq 12, \\ x_1 - x_2 &\geq 0, \\ 0 &\leq x_1 \leq 20. \end{aligned}$$

[P.U.B.E. (Mech.) May, 1983; P.T.U. (B.Tech.) 2001; Nellore M.B.A. 2002]

Solution

The solution space satisfying the given constraints and meeting the non-negativity restrictions $x_1 \geq 0$ and $x_2 \geq 0$ is shown shaded in Fig. 2.4. Any point in this shaded region is a feasible solution to the given problem.

The co-ordinates of the five vertices of the convex region ABCDE are A(3, 3), B(12, 12), C(18, 12), D(20, 10) and E(20, 3).

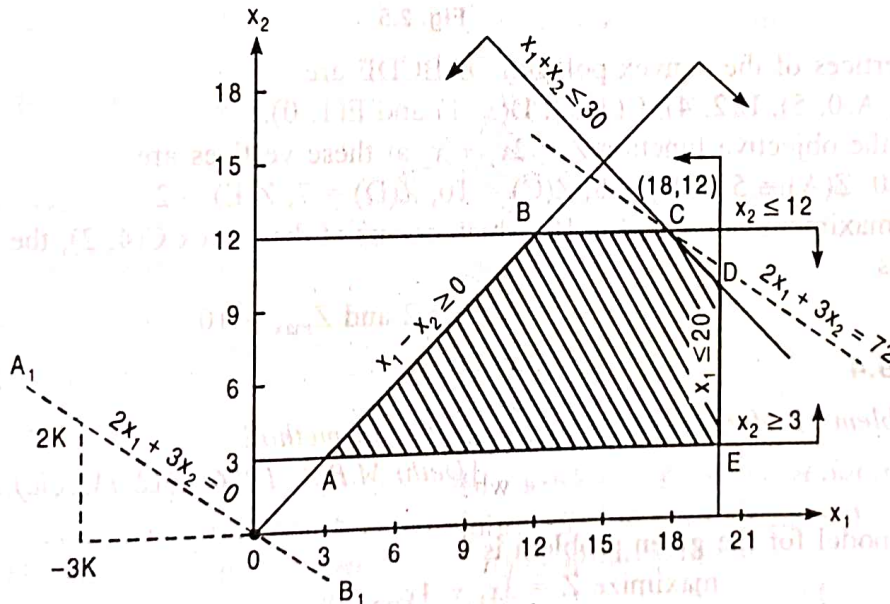


Fig. 2.4

Values of the objective function $Z = 2x_1 + 3x_2$ at these vertices are $Z(A) = 15$, $Z(B) = 60$, $Z(C) = 72$, $Z(D) = 70$ and $Z(E) = 49$.

Since the maximum value of Z is 72, which occurs at the point C(18, 12), the solution to the given problem is $x_1 = 18$, $x_2 = 12$, $Z_{\max} = 72$.

EXAMPLE 2.9.3

$$\begin{aligned} \text{Maximize } Z &= 2x_1 + x_2, \\ \text{subject to } x_1 + 2x_2 &\leq 10, \\ x_1 + x_2 &\leq 6, \end{aligned}$$

$$x_1 - x_2 \leq 2,$$

$$x_1 - 2x_2 \leq 1,$$

$$x_1, x_2 \geq 0.$$

[P.U.B.E. (E. and Ec.) 1999; B.E. (Elect.) Oct., 1993; B.Com. Sept., 2004]

Solution

The solution space satisfying the given constraints and meeting the non-negativity restrictions $x_1 \geq 0$ and $x_2 \geq 0$ is shown shaded in Fig. 2.5. Any point in this shaded region is a feasible solution to the given problem.

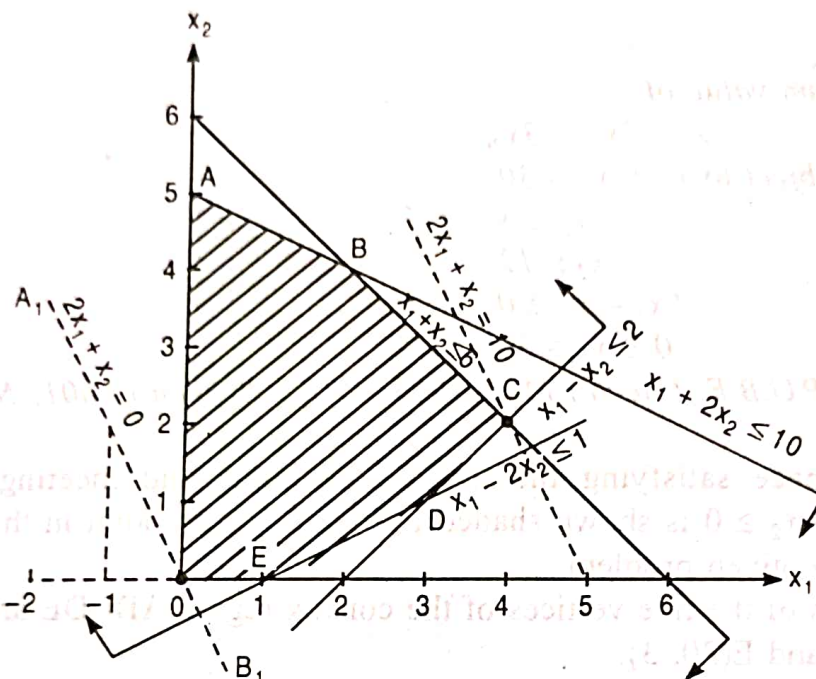


Fig. 2.5

The six vertices of the convex polygon OABCDE are

O(0, 0), A(0, 5), B(2, 4), C(4, 2), D(3, 1) and E(1, 0).

Values of the objective function $Z = 2x_1 + x_2$ at these vertices are

$Z(O) = 0$, $Z(A) = 5$, $Z(B) = 8$, $Z(C) = 10$, $Z(D) = 7$, $Z(E) = 2$.

Since the maximum value of Z is 10, which occurs at the vertex C(4, 2), the solution to the given problem is

$$x_1 = 4, x_2 = 2 \text{ and } Z_{\max} = 10.$$

EXAMPLE 2.9.4

Solve problem 12 of exercise 2.1 by the graphical method.

[Delhi M.B.A. 1976; P.U. (Mech.) August, 1978]

Solution

The L.P. model for the given problem is

$$\text{maximize } Z = 3x_1 + 4x_2,$$

$$\text{subject to } 5x_1 + 4x_2 \leq 200,$$

$$3x_1 + 5x_2 \leq 150,$$

$$5x_1 + 4x_2 \geq 100,$$

$$8x_1 + 4x_2 \geq 80,$$

$$x_1, x_2 \geq 0.$$

The solution space satisfying the given constraints and meeting the non-negativity restrictions $x_1 \geq 0$ and $x_2 \geq 0$ is shown shaded in Fig. 2.6.

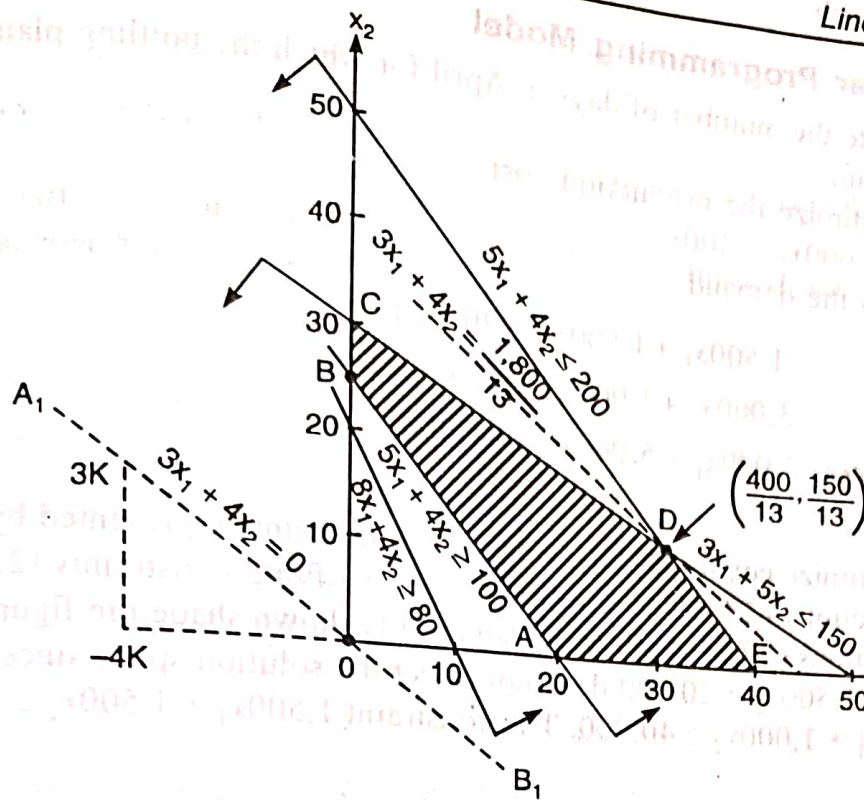


Fig. 2.6

Any point in the shaded region is a feasible solution to the given problem. The co-ordinates of the vertices of this convex region ABCDE are

$$A(20, 0), B(0, 25), C(0, 30), D\left(\frac{400}{13}, \frac{150}{13}\right) \text{ and } E(40, 0).$$

Values of the objective function $Z = 3x_1 + 4x_2$ at these vertices are

$$Z(A) = 60, Z(B) = 100, Z(C) = 120, Z(D) = \frac{1,800}{13}, Z(E) = 120.$$

Since the maximum value of Z is $\frac{1,800}{13}$ which occurs at the vertex $D\left(\frac{400}{13}, \frac{150}{13}\right)$, the solution to the given problem is

$$x_1 = \frac{400}{13}, x_2 = \frac{150}{13} \text{ and } Z_{\max} = \text{Rs. } \frac{1,800}{13} = \text{Rs. } 138.46.$$

EXAMPLE 2.9.5

Mohan-Meakins Breveries Ltd. has two bottling plants, one located at Solan and the other at Mohan Nagar. Each plant produces three drinks, whisky, beer and fruit juices named A, B and C respectively. The number of bottles produced per day are as follows:

	Solan (S)	Mohan Nagar (M)
Whisky, A	1,500	1,500
Beer, B	3,000	1,000
Fruit juices, C	2,000	5,000

A market survey indicates that during the month of April, there will be a demand of 20,000 bottles of whisky, 40,000 bottles of beer and 44,000 bottles of fruit juices. The operating costs per day for plants at Solan and Mohan Nagar are 600 and 400 monetary units. For how many days each plant be run in April so as to minimize the production cost, while still meeting the market demand?

[P.U.B.E. (Elect.) 2001, I.A.S. 1987]

Formulation of Linear Programming Model

Let x_1 and x_2 denote the number of days in April for which the bottling plants at Solan and Mohan Nagar must be run.

Objective is to minimize the production cost.

i.e., minimize $Z = 600x_1 + 400x_2$.

Constraints are on the demand.

$$\left. \begin{array}{l} \text{i.e., for whisky, } 1,500x_1 + 1,500x_2 \geq 20,000, \\ \text{for beer, } 3,000x_1 + 1,000x_2 \geq 40,000, \\ \text{and for fruit juices, } 2,000x_1 + 5,000x_2 \geq 44,000, \end{array} \right\}$$

where $x_1, x_2 \geq 0$.

Thus we are to optimize equation (2.4), subject to constraints represented by relations (2.5) and nonnegativity restrictions (2.6). The solution space satisfying constraints (2.5) and meeting the non-negativity conditions (2.6) (which is unbounded) is shown shaded in figure 2.7. Note that the constraint $1,500x_1 + 1,500x_2 \geq 20,000$ does not affect the solution space since it is dominated by the constraint $3,000x_1 + 1,000x_2 \geq 40,000$. The constraint $1,500x_1 + 1,500x_2 \geq 20,000$ is called a *redundant constraint*.

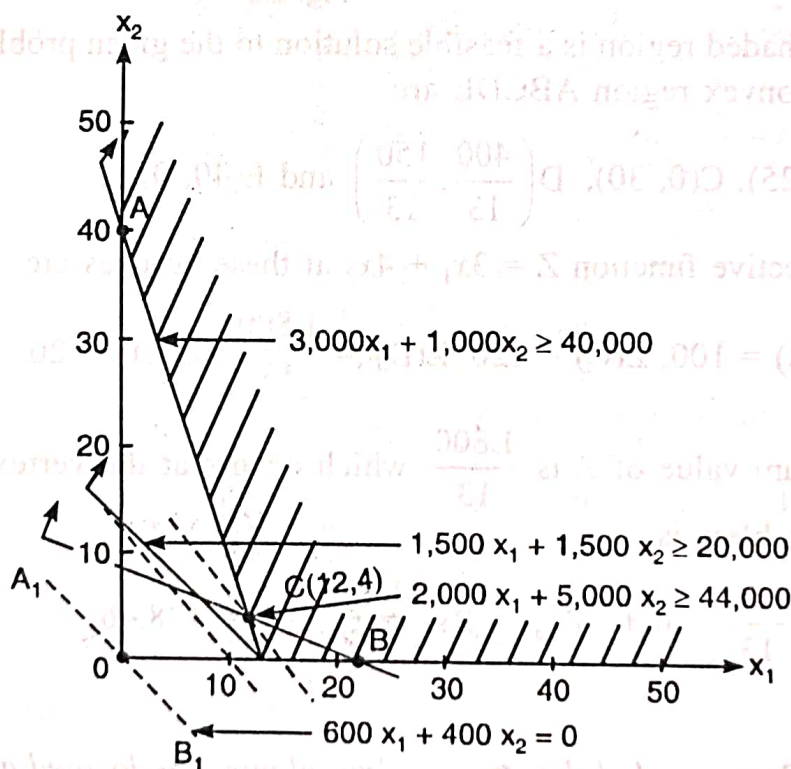


Fig. 2.7

To Find Optimal Solution

Method 1. The objective function, when $Z = 0$, gives the equation $600x_1 + 400x_2 = 0$ which is shown by the dotted line $A_1 B_1$ passing through the origin O. As Z is increased from zero, the dotted line $A_1 B_1$ moves to the right, parallel to itself. Since we are interested in minimizing Z , we increase the value of Z till the dotted line passes through the *nearest corner* of the shaded region from the origin. This gives the minimum value of Z , while keeping x_1 and x_2 within the region of feasible solutions. The coordinates of this point C are (12, 4). Thus production cost will be minimum if plants at Solan and Mohan Nagar are run for 12 days and 4 days respectively.

Method 2. The three vertices of the convex set ACB are $A(0, 40)$, $B(22, 0)$ and $C(12, 4)$. Values of the objective function $Z = 600x_1 + 400x_2$ at these vertices are $Z(A) = 16,000$, $Z(B) = 13,200$ and $Z(C) = 7,200 + 1,600 = 8,800$. Thus the minimum value of Z is 8,800 monetary units and it occurs at the vertex $C(12, 4)$. Hence the solution to the problem is

$$x_1 = 12 \text{ days,}$$

$$x_2 = 4 \text{ days,}$$

$$Z_{\min} = 8,800 \text{ monetary units.}$$

EXAMPLE 2.9.6

Find the minimum value of

$$\begin{aligned} Z &= 5x_1 - 2x_2, \\ 2x_1 + 3x_2 &\geq 1, \\ x_1, x_2 &\geq 0. \end{aligned}$$

Solution

The solution space satisfying the given constraint and meeting the non-negativity restrictions $x_1 \geq 0$ and $x_2 \geq 0$ is shown shaded in Fig. 2.8. Any point in the shaded region is a feasible solution to the given problem.

[Madurai B.E. (Mech.) 1976]

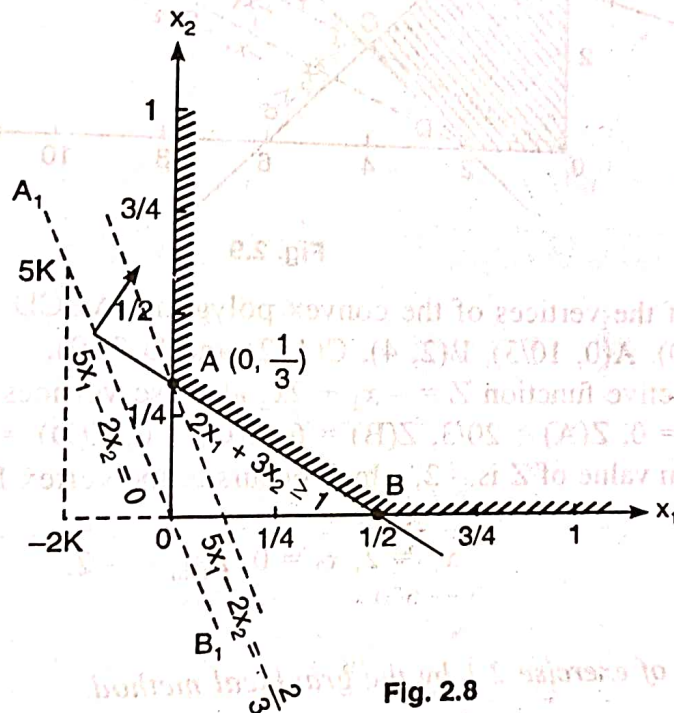


Fig. 2.8

The coordinates of the two vertices of the unbounded convex region are

$$A\left(0, \frac{1}{3}\right) \text{ and } B\left(\frac{1}{2}, 0\right).$$

Values of the objective function $Z = 5x_1 - 2x_2$ at these vertices are

$$Z(A) = -\frac{2}{3}, \quad Z(B) = \frac{5}{2}.$$

Since the minimum value of Z is $-2/3$ which occurs at the vertex $A(0, 1/3)$, the solution to the given problem is

$$x_1 = 0, \quad x_2 = \frac{1}{3}, \quad Z_{\min} = -\frac{2}{3}.$$

EXERCISES 2.2

1. What are the advantages of linear programming approach ? State the limitations of L.P.
[P.U.B.E. (E. & Ec.) 1999; B.E. (T. & I.T.) Nov., 2004 ;
Kuru. U. M. Tech. May, 1988; G.N.D.U.B. Com. 1995]
2. Explain the 'graphic method' of solving an L.P.P. What are its limitations ?
3. How will you solve an L.P.P. graphically ? Give examples using graph. [G.N.D.U. B.Com. 1995]
4. Where is the optimal solution of an L.P.P. located on a graph ? What is the effect of changing the objective function on the optimum point ? Discuss with examples.
5. Why do some problems have multiple optimal feasible solutions ? How such information is useful for decision making ? Explain.
6. Explain with example an L.P. problem which has no feasible solution. Use graphical method to explain.
[P.U.B.E. (Mech.) 2000]
7. Explain the following concepts in the context of linear programming:
 - (i) Objective function
 - (ii) Convex polygon
 - (iii) Redundant constraint[IGNOU M.C.A. 2003 ; GNDU B.Com. Sept., 1991]

Use graphical method to solve the following problems:

8. Maximize $Z = 2x_1 + 3x_2$,
subject to constraints $x_1 + x_2 \leq 1$,

$$3x_1 + x_2 \leq 4,$$

$$x_1, x_2 \geq 0.$$

[Meerut B.Sc. (Math.) 1970]
(Ans. $x_1 = 0, x_2 = 1; Z_{\max} = 3$.)

9. Maximize
subject to

$$\begin{aligned} Z &= 5x_1 + 3x_2, \\ 3x_1 + 5x_2 &\leq 15, \\ 5x_1 + 2x_2 &\leq 10, \\ x_1, x_2 &\geq 0. \end{aligned}$$

[Delhi M.Sc. (Math.) 1969]

$$\left(\text{Ans. } x_1 = \frac{20}{19}, x_2 = \frac{45}{19}; Z_{\max} = \frac{235}{19} \right)$$

10. Maximize
subject to

$$\begin{aligned} Z &= 5x_1 + 3x_2, \\ x_1 + x_2 &\leq 6, \\ 0 \leq x_1 &\leq 3, \\ 0 \leq x_2 &\leq 3, \\ 2x_1 + 3x_2 &\geq 3. \end{aligned}$$

(Ans. $x_1 = 3, x_2 = 3; Z_{\max} = 24$.)

11. Minimize
subject to

$$\begin{aligned} Z &= 20x_1 + 10x_2, \\ x_1 + 2x_2 &\leq 40, \\ 3x_1 + x_2 &\geq 30, \\ 4x_1 + 3x_2 &\geq 60, \\ x_1, x_2 &\geq 0. \end{aligned}$$

[P.U.M.B.A. 2003; Meerut M.Com. 1977]
(Ans. $x_1 = 6, x_2 = 12; Z_{\min} = 240$.)

12. Maximize
subject to

$$\begin{aligned} Z &= x_1 + \frac{x_2}{2}, \\ 3x_1 + 2x_2 &\leq 12, \\ 5x_1 &\leq 10, \\ x_1 + x_2 &\leq 8, \\ -x_1 + x_2 &\geq 4, \\ x_1, x_2 &\geq 0. \end{aligned}$$

[Pb. Univ. B.E. (Mech.) Engg. Nov., 1979]
(Ans. $x_1 = 0.8, x_2 = 4.8; Z_{\max} = 3.2$.)

13. Minimize
subject to

$$\begin{aligned} Z &= 60x + 40y, \\ 30x + 10y &\geq 240, \\ 10x + 10y &\geq 160, \\ 20x + 60y &\geq 480, \\ x, y &\geq 0. \end{aligned}$$

[Pb. Univ. B. Com. 2006; B.E. (Mech.) Engg. Nov., 1980]
(Ans. $x = 6, y = 6; Z_{\min} = 600$.)

14. Minimize
subject to

$$\begin{aligned} Z &= 4x_1 + x_2, \\ 3x_1 + 4x_2 &\geq 20, \\ -x_1 - 5x_2 &\leq -15, \\ x_1, x_2 &\geq 0. \end{aligned}$$

[P.U.B. Com. 2006]
(Ans. $x_1 = 0, x_2 = 5; Z_{\min} = 5$.)

15. Minimize
subject to

$$\begin{aligned} Z &= 3x_1 + 2x_2, \\ 8x_1 + x_2 &\geq 8, \\ 2x_1 + x_2 &\geq 6, \\ x_1 + 3x_2 &\geq 6, \\ x_1 + 6x_2 &\geq 8, \\ x_1, x_2 &\geq 0. \end{aligned}$$

[Pb. Univ. B.E. (Mech.) Dec., 1985]
(Ans. $x_1 = 2.4, x_2 = 1.2; Z_{\min} = 9.6$.)

16. Maximize
subject to
- $$Z = 3x + 9y,$$
- $$x + 4y \leq 8,$$
- $$x + 2y \leq 4,$$
- $$x, y \geq 0.$$
- [Nagpur Univ. B.E., 2003]
(Ans. $x = 0, y = 2; Z_{\max} = 18.$)
17. Maximize
subject to
- $$Z = 6x_1 + 4x_2,$$
- $$2x_1 + 3x_2 \geq 30,$$
- $$3x_1 + 2x_2 \leq 24,$$
- $$x_1 + x_2 \geq 3,$$
- $$x_1, x_2 \geq 0.$$
- [P.U.B.Com. 2002]
(Ans. $x_1 = 12/5, x_2 = 42/5; Z_{\max} = 48.$)
18. Maximize
subject to
- $$Z = 2x_1 + x_2,$$
- $$x_1 + 2x_2 \leq 10,$$
- $$x_1 + x_2 \leq 6,$$
- $$x_1 - x_2 \leq 2,$$
- $$x_1 - 2x_2 \leq 1,$$
- $$x_1, x_2 \geq 0$$
- [P.U.B.Com. 2002]
(Ans. $x_1 = 4, x_2 = 2; Z_{\max} = 10.$)
19. Minimize
subject to
- $$Z = 5x_1 + 6x_2,$$
- $$2x_1 + 5x_2 \geq 1,500,$$
- $$3x_1 + x_2 \geq 1,200,$$
- $$x_1, x_2 \geq 0.$$
- where
- [P.U.B.B.A. 2001; B.E. (Mech.) 1997]
(Ans. $x_1 = \frac{4,500}{13}, x_2 = \frac{2,100}{13}; Z_{\min} = \frac{35,100}{13}.$)
20. Maximize
subject to
- $$Z = -150x_1 - 100x_2 + 2,80,000,$$
- $$20 \leq x_1 \leq 60,$$
- $$70 \leq x_2 \leq 140,$$
- $$120 \leq x_1 + x_2 \leq 140,$$
- $$x_1, x_2 \geq 0.$$
- [P.U.B. Com. 2000]
(Ans. $x_1 = 20, x_2 = 100, Z_{\max} = 2,67,000.$)
21. Maximize
subject to
- $$Z = 3x_1 + 2x_2,$$
- $$2x_1 + x_2 \leq 10,$$
- $$x_1 + 3x_2 \leq 6,$$
- $$x_1, x_2 \geq 0.$$
- [P.U.B. Com. 1999]
(Ans. $x_1 = \frac{24}{5}, x_2 = \frac{2}{5}; Z_{\max} = \frac{76}{5}.$)
22. Maximize
subject to
- $$Z = 3x_1 + 2x_2,$$
- $$-2x_1 + x_2 \leq 1,$$
- $$x_1 \leq 2,$$
- $$x_1 + x_2 \leq 3,$$
- $$x_1, x_2 \geq 0.$$
- [Pbi.U. MCA, 2001]
(Ans. $x_1 = 2, x_2 = 1; Z_{\max} = 8.$)
23. (a) Minimize
(b) Maximize
constraints are
- $$Z = 2x_1 + x_2,$$
- $$Z = 2x_1 + 3x_2,$$
- $$x_1 - 3x_2 \leq 6,$$
- $$2x_1 + 4x_2 \geq 8,$$

$$x_1 - 3x_2 \geq -6,$$

$$x_1, x_2 \geq 0.$$

[P.U.B.E. (Elect.) 2001, 1999]

(Ans. (a) $x_1 = 0, x_2 = 2$; $Z_{\min} = 2$, (b) Unbounded.)

24. Maximize
subject to

$$Z = 80x_1 + 120x_2,$$

$$x_1 + x_2 \leq 9,$$

$$20x_1 + 50x_2 \leq 360,$$

$$x_1 \geq 2,$$

$$x_2 \geq 3,$$

$$x_1, x_2 \geq 0.$$

[P.U.B.Com. April, 2003]

(Ans. $x_1 = 3, x_2 = 6$; $Z_{\max} = 960$.)

25. Consider the following L.P.P. graphically:

Maximize
subject to

$$Z = 50x_1 + 60x_2,$$

$$2x_1 + 3x_2 \leq 1,500,$$

$$3x_1 + 2x_2 \leq 1,500,$$

$$x_1 \leq 400,$$

$$x_2 \leq 400,$$

$$2x_1 \leq 3x_2,$$

$$x_1, x_2 \geq 0.$$

(i) Construct the constraint lines on the graph.

(ii) Mark the feasible region area.

(iii) Graph at least two isoprofit lines and mark direction of improvement of Z .

(iv) Obtain the optimal solution for the problem.

[Karn. U.B.E. (Mech.) 1995]

(Ans. (iv) $x_1 = 300, x_2 = 300$; $Z_{\max} = 33,000$.)

26. Maximize
subject to

$$Z = 5x_2 - x_1,$$

$$2x_1 - x_2 \geq -2,$$

$$-0.2x_1 + x_2 \leq 2,$$

$$x_1, x_2 \geq 0.$$

[Pb. Univ. B.E. (Mech.) Engg. Nov., 1981]

(Ans. x_1, x_2 can be arbitrarily large; $Z_{\max} = 10$.)

27. Maximize
subject to

$$Z = x_1 + 0.5x_2,$$

$$3x_1 + 2x_2 \leq 6,$$

$$x_1 \leq 2,$$

$$x_1 + x_2 \leq 8,$$

$$-x_1 + x_2 \geq 4,$$

$$x_1, x_2 \geq 0.$$

[Pb. Univ. B.E. (Mech.) Engg. May, 1982]

(Ans. Infeasible solution.)

28. Maximize

$$Z = 2x_1 + x_2,$$

subject to

$$\frac{3}{2}x_1 + x_2 \leq 6,$$

$$x_1 \leq 2,$$

$$x_1 + x_2 \geq 7,$$

$$-x_1 + x_2 \geq 4,$$

$$x_1, x_2 \geq 0.$$

and

(Ans. Solution does not exist.)

29. Maximize
subject to

$$Z = 3x_1 + 4x_2,$$

$$x_1 - x_2 \leq -1,$$

$$x_1 \geq x_2,$$

$$x_1, x_2 \geq 0.$$

[Meerut M.Sc. (Math.) 1974]

(Ans. Solution does not exist.)

30. Maximize $Z = 8x_1 + x_2$,
 subject to $8x_1 + x_2 \leq 8$,
 $2x_1 + x_2 \leq 6$,
 $3x_1 + x_2 \leq 6$,
 $x_1 + 6x_2 \leq 8$,
 $x_1, x_2 \geq 0$.

(Ans. Infinite number of optimal solutions exist.)

31. Maximize $Z = 4x_1 + 5x_2$,
 subject to $x_1 + x_2 \geq 1$,
 $-2x_1 + x_2 \leq 1$,
 $4x_1 - x_2 \geq 1$,
 $x_1, x_2 \geq 0$.

[R.E.C. Hamirpur, 1995]

(Ans. Unbounded solution.)

32. Find the maximum as well as minimum value of the objective function

$$\begin{aligned} Z &= 4x + 5y, \\ \text{subject to } 2x + y &\leq 6, \\ x + 2y &\leq 5, \\ x - 2y &\leq 2, \\ -x + y &\leq 2, \\ x + y &\geq 1, \\ \text{and } x, y &\geq 0. \end{aligned}$$

[P.U.B.E. (E. & Ec.) 1999; B.E. (Elect.) Oct., 1993; NIIFT Mohali, 1999]

$$\left(\text{Ans. } x = \frac{7}{3}, y = \frac{4}{3}, Z_{\max} = 16; x = 1, y = 0, Z_{\min} = 4. \right)$$

33. Find the maximum as well as minimum value of the objective function

$$\begin{aligned} Z &= -3x + 6y, \\ \text{subject to } x + 2y + 1 &\geq 0, \\ -4x + y + 23 &\geq 0, \\ 2x + y - 4 &\geq 0, \\ x - 4y + 13 &\geq 0, \\ x - y + 1 &\geq 0, \\ \text{and } x &\geq 0, y \geq 0. \end{aligned}$$

$$\left(\text{Ans. } x = 3, y = 4, Z_{\max} = 15; x = 5.75, y = 0; Z_{\min} = 17.25. \right)$$

34. To products A and B are to be manufactured. One unit of product A requires 2.4 minutes of punch press time and 5 minutes of assembly time. The profit for product A is Re. 0.60 per unit. One unit of product B requires 3 minutes of punch press time and 2.5 minutes of welding time. The profit for product B is Re. 0.70 per unit. The capacity of the punch press deptt. available for these products is 1,200 minutes/week. The welding deptt. has an idle capacity of 600 minutes/week and assembly deptt. has 1,500 minutes/week.

(i) Formulate the problem as linear programming problem.

(ii) Determine the quantities of products A and B so that total profit is maximized.

[Pb. Univ. (Mech.) Engg. April, 1976]

$$\left(\text{Ans. } x_A = 300, x_B = 160 \text{ units; } Z_{\max} = \text{Rs. } 292. \right)$$

35. A feed mixing operation can be described in terms of two activities. The required mixture must contain four kinds of ingredients w, x, y and z. Two basic feeds A and B, which contain the required ingredients are available in the market. 1 kg of A contains 0.1 kg of w, 0.1 kg of y and 0.2 kg of z. Likewise, 1 kg of feed B contains 0.1 kg of x, 0.2 kg of y and 0.1 kg of z. The daily per head requirement is of at least, 0.4 kg of w, 0.6 kg of x, 2 kg of y and 1.6 kg of z. Feed A can be bought for £0.07 per kg and B for £0.05 per kg. Determine the quantity of feeds A and B in the mixture in order that the total cost is minimum.

$$\left(\text{Ans. } x_A = \frac{16}{3}, x_B = \frac{22}{3}; Z_{\min} = \text{£ } 0.74. \right)$$